

Indian TTS Dataset Curation Report

Candidate: Lochana Balivada
Deadline: Sunday, 7 June 2026, 11:59 PM IST

Dataset Links

- Hugging Face dataset: <https://huggingface.co/datasets/lochanab30/indian-tts-60min>
- GitHub repository: <https://github.com/lochana-b/sarvam-assignment-indian-tts-dataset-curation>

Summary

I built a 60-minute TTS training dataset from YouTube-sourced clean single-speaker speech:

Bucket	Language	Target	Final duration	Samples
Indian English	en-IN	30 min	30.17 min	32
Hindi	hi-IN	30 min	29.48 min	31
Total	Mixed	60 min	59.65 min	63

The dataset achieved near-perfect balance with an 87.5% acceptance rate (63 kept from 72 candidates), demonstrating quality-first curation.

Pipeline

The pipeline is manifest-driven:

1. I listed candidate YouTube sources and timestamps in `config/sources.yaml`.
2. `tts-curate download` downloaded source audio locally.
3. `tts-curate cut` extracted candidate segments, converted them to 16 kHz mono WAV, and applied light loudness normalization.
4. `tts-curate asr` used Sarvam Saaras v3 batch STT with diarization enabled.
5. `tts-curate tag` used a Sarvam chat model to label emotion and style.
6. `tts-curate review-sheet` generated a CSV for manual listening review.
7. I manually listened, corrected transcripts, rejected bad clips, and marked approved clips as `keep`.

8. `tts-curate finalize` created final metadata.
9. `tts-curate publish` pushed the reviewed dataset to Hugging Face.

Source Selection

Describe the channels/videos you used, why you selected them, and how you checked that each had a clean single-speaker region.

Source	Language	Reason chosen	Rejected portions
TEDx Talk (Pooja Jambotkar)	Indian English	Professional presentation, clear audio, single speaker, formal style	1 seg transcript wasn't generated properly
TEDx Talk (Amina Nijam)	Indian English	Different speaker for variety, emotional storytelling, good prosody	None - high quality throughout
Palki Sharma Speech	Indian English	News anchor quality, passionate delivery, professional recording	Clapping/applause (tried to avoid)
Arnab Goswami Debate	Indian English	Solo commentary only, energetic/angry emotion for variety, news style	Panel discussion portions (not included in segments)
Hindi Astrology (Astro Arun Pandit)	Hindi	Calm/devotional style, 12 natural segments (rashis), single speaker	Intro/outro music avoided
Hindi Spiritual Story	Hindi	Narrative style, emotional variety, devotional content	None - clean narration
Hindi Story (Moral Stories)	Hindi	Additional narrative content, different speaker, clear enunciation	None - consistent quality

Selection process: Previewed each video at 1.5x speed to identify 30-60 second single-speaker sections, avoiding intro/outro music, audience noise, and multi-speaker segments.

Iterations

Document concrete improvements you made. Examples:

Iteration	Problem found	Change made	Result
1	Intro music was present in early clips	Moved start timestamps past intros	Cleaner speech-only clips
2	Diarization showed two speakers in interview answers	Kept monologue stretches only	Better single-speaker consistency
3	ASR normalized some spoken numbers incorrectly	Manually corrected transcripts	More faithful TTS text

Quality Decisions

Summarize your judgment calls:

- Minimum acceptable audio quality: Manually listened to every audio and selected with care.
- How you decided emotion/style tags: As I was listening to the audio, I was able to judge the style/emotion.
- How you handled unclear words: While manually reviewing transcripts, I corrected the transcripts.

What Worked

- Pipeline automation: Script took care of the entire pipeline
- Sarvam API quality: **Saaras v3 STT**:
- High quality transcriptions (estimated 80-85% accuracy for clear audio)
- Diarization correctly identified single speakers
- Handled both Indian English and Hindi well
- API was reliable with batch processing

Chat Completion API: - Attempted for emotion/style tagging - Encountered timeouts/failures - Manual tagging proved more accurate anyway

- Manual review process: Manually listened to every audio clipping and checked if transcripts were correct or not. If incorrect, manually fixed the transcripts

What Did Not Work

- Some transcripts were generated as empty.

- ASR was putting punctuation marks incorrectly.
- Due to time constraints I couldn't experiment more with Sarvam API's.
- Attempted Sarvam Chat Completion (Emotion tagging), but failed with "API ERROR"

Observations

ASR Strengths/Weaknesses:

Indian English: - Generally accurate for clear speakers (estimated 80-85% accuracy) - Struggled with proper nouns and technical terms - Grammar artifacts: repeated words ("I'm still I am"), unnecessary conjunctions ("And but") - Punctuation needed frequent correction - Numbers in spoken form sometimes incorrect

Hindi: - Comparable accuracy to English - Diacritic marks sometimes incorrect - Handled devotional and narrative content well

Diarization Reliability:

Mostly reliable - correctly identified single speaker (count=1) in final 63 segments. Few false positives where pauses were flagged as speaker changes.

Common Rejection Causes (9 rejected):

- Background music/noise: 3 segments
- Multiple speakers: 2 segments
- Empty transcripts: 2 segments
- Poor audio quality/clipping: 2 segments

Transcript Correction Patterns:

41/63 transcripts (65%) corrected. Common patterns: - Grammar and stutters: "I'm still I am" → "I still am" - Punctuation: adding commas, periods, proper capitalization - Minor word changes: "at some point" → "in some point" - Proper nouns and technical terms requiring correction

Improvements With More Time

1. **Better data collection:** I would take better videos with more diverse speakers and content types.

2. **More speakers:** Limited diversity (some speakers in multiple segments). Would add 5-10 more speakers per language.
3. **Stricter licensing:** Contact creators directly for explicit permission.
4. **Phonetic balance:** Analyze phoneme distribution for complete coverage.
5. **Objective metrics:** Use SNR/PESQ metrics alongside subjective assessment.
6. **Second-pass review:** Have another person review samples for consistency.
7. **Train/dev/test split:** Create 80/10/10 with speaker-level separation.
8. **Richer prosody labels:** Add speaking rate, pitch range, pause patterns.
9. **Regional diversity:** Add accent variety within languages.
10. **Longer segments:** Target 40-60s sweet spot (some are at 20s minimum).
11. **Quality automation:** Build automated checks for common issues to pre-filter candidates.